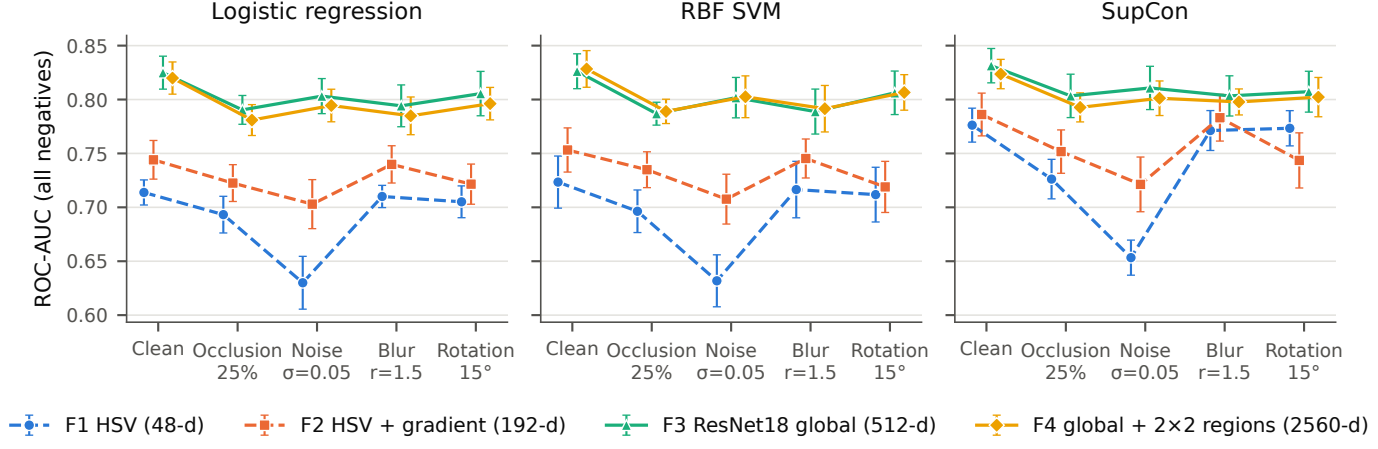
Logistic regression
RBF SVM
SupCon
ROC-AUC (all negatives)
0.85
0.80
0.75
0.70
0.65
0.60
Clean
Occlusion 25%
Noise σ=0.05
Blur r=1.5
Rotation 15°
F1 HSV (48-d)
F2 HSV + gradient (192-d)
F3 ResNet18 global (512-d)
F4 global + 2×2 regions (2560-d)